

SOP 1 - Setting Up a Main Assembly & Its CPQ Flow

Who this is for: the person who builds a product so it can be configured, rendered, and priced in the CPQ. **Examples used throughout:** **Flat Iron** (M2C brand) and **Brimar FR / Brimar Signature** (Classical Elements brand) - both straight-pole drapery hardware. **Focus:** setup -> node grouping -> generate the flow -> edit the steps -> **pricing & customer pricing**.

Tip: switch to the correct **brand** first (top of HQ). Flat Iron lives in M2C; Brimar lives in Classical Elements.

Before you start - what makes an assembly a "Main Assembly"

An assembly only appears in Node Grouping / the flow tools when **both** are true:

1. **It is flagged MAIN.** In **Tab 1.4 BOM Engine**, the assembly's **"Routing Classification"** dropdown must be set to **MAIN** (not Unassigned/Standard). This is what marks it as the top-level product that drives a CPQ flow.
2. **It has a 3D model (.glb).** Upload the model in **Tab 1 Inception** (the file picker accepts .glb, or use **"Upload Revision"**). It is stored on the assembly as its CAD model. Without it, Node Grouping shows *"Please upload a 3D CAD model to manage nodes."*

So: Flat Iron and Brimar must each be Routing Classification = **MAIN** and have their .glb uploaded before the steps below.

Part A - Node Grouping (Tab 1.5): tag the model so it can render

This is where you teach the model which meshes are brackets, poles, finials, etc., and where they sit (left/right, wall/ceiling). Everything the CPQ shows or hides depends on this.

A1. Select the assembly

- Use the **Search / Project / Status** filters on the left. Status chips read **Awaiting CAD**, **Needs Clustering**, or **Clustered (n)**.
- Click the assembly card (e.g. **Flat Iron**). The 3D model loads in the center viewer. Camera tip: *right-click & drag to pan.*

A2. Auto-Group the parts

1. Click **** Auto-Group (n)**** (top-right of the viewer; n = parts detected). This opens the **"Group by Location & Position"** panel.
2. Choose **Split by Position** (one cluster per physical instance, each labeled LEFT/CENTER/RIGHT) - this is what you want for hardware. (*Merge Instances* lumps all copies into one; use only for parts that are truly one group.)
3. **** Flip L/R** - this is the Brimar fix.** Flat Iron's model is built so screen-left = LEFT automatically. Brimar is mirrored, so its left/right come out swapped - toggle **Flip L/R** once and the labels correct for the whole assembly. (Setting is remembered per assembly.)
4. Click **Select all**, then **** From Library**** - this auto-fills each group's **Location** and **Category** from the matched library part (the most reliable tagging). You can also bulk-tag with the **WALL**

/ **CEILING** / **END** and **BRACKET** / **POLE** / **FINIAL** / **BACKPLATE** / **RING** buttons under "Tag checked:".

5. Review each row's **Loc:** and **Cat:** tags; fix any that are wrong with the per-row buttons. Use **Edit** on a row to click meshes in 3D and add/remove them from that group.
6. Click **Save Clusters**.

A3. Verify with Locate (the "glow")

- In the **Saved BOM Bindings** list, click **Locate** on a cluster - it lights that cluster's meshes gold in 3D and fades everything else. This is how you confirm a cluster contains *exactly* the right meshes (e.g. the center bracket is really the center bracket). If it lights up extra/foreign parts, fix it (re-Edit or split) before generating a flow.
- **Highlight Unassigned** (viewer toolbar) glows any meshes not yet in a cluster red - use it to catch leftovers.

A4. Fix a mislabeled cluster anytime (no re-run needed)

Each saved cluster card has **WALL/CEILING/END**, **LEFT/CENTER/RIGHT**, and category buttons. Click the right one to correct it - only that tag changes, the meshes stay. (**Clear all** wipes the assembly's clusters to start over.)

What the tags do downstream

- **Category** (BRACKET/POLE/FINIAL/BACKPLATE/RING) - what the part is; the flow generator builds a step per category.
- **Location** (WALL/CEILING/END) - the mount context; controls which parts show for a given mount.
- **Position** (LEFT/CENTER/RIGHT) - which physical spot; CENTER is what the "passing bracket" clone uses.

Part B - Generate the CPQ flow (Tab 11 System Admin -> "CPQ Flow Builder")

1. Open **Tab 11 System Admin -> CPQ Flow Builder**.
2. Under "**Active CPQ Flows**", find the "- assembly to generate from -" dropdown, pick your assembly (e.g. Flat Iron), and click **Generate Flow from Tags**.
3. This reads your clusters + pins and creates one flow with a standard step set: **Pole / Rod Material, Pole Length & Finish**, per-position **Bracket & Mount** steps (center marked "clone along pole"), per-end **End Treatment, Rings, Splice, Cut / Splice Fee**.

One flow = one configuration. The builder is **1:1 with a bay configuration + end style + projection** (the "Fabrication Preset" block). So you generate a separate flow for each mount/end combination you sell (e.g. Flat Iron - Wall - Finial; Flat Iron - Ceiling - Miter). Name them clearly.

Part C - Edit the flow & its steps

Click a flow card to load it. You'll edit two levels: the whole flow, then each step.

Flow-level settings

- **Link to Master Assembly** - ties the flow to the 3D model (must match the assembly you grouped). This is what makes it render.
- **CPQ Flow Name, ERP Item ID, Base Price (\$)** - the flow's identity and starting price.
- **Fabrication Preset - Bay Configuration / End Style / Bracket Projection** - stamps what config this flow represents (these auto-seed the Vision tool later).
- **Default Finishes (cascade to every step)** + ****Apply to all steps**** - set the finish options once for the whole flow.
- **Hide Geometry (other configs)** - force-hide clusters this configuration never shows.
- **Save and Cascade to Master** when done.

The step editor

Each step row shows **Step N:**, its Type, Data, and Required flag, with **(up)/(down)** to reorder, **Edit**, and **Del**. In the editor, the key fields:

- **Title** - what the customer sees ("Select Bracket Style").
- **Type** - choose how the step behaves:
 - **Choose / Swap Style (STYLE_SWAP)** - curated BOM items the customer swaps between; each option carries its own price and 3D mesh. Use for brackets, poles, finials.
 - **Dropdown List (DROPDOWN)** - pick one from a data source (e.g. finishes, rings).
 - **Visual Grid / Visual Grid + Dimensions** - image/swatch picker (+ a size calculator).
 - **Dimensional Input Only (DIMENSIONS)** - a measurement, priced by a calculator template.
 - **Static Fee / Quantity (STATIC_FEE)** - a flat fee $\times$ quantity (splices, cut fees).
- **Clone along pole (center passing bracket)** - check on the center bracket step; the chosen bracket is cloned by the step quantity and spaced down the pole in 3D.
- **Tag-driven Mount selector + Applies to position** - a step that shows only the parts matching the chosen mount (Wall/Ceiling/Inside).
- **Restrict Options** - limit which library items/finishes this step offers.
- **Style Options (STYLE_SWAP)** - tick which BOM items can be swapped; each gets a **price**, a **Z layer**, and **Proj**". Optionally **"Also let the customer pick a Finish for the chosen style."**
- **Geometry Swap (Mesh) / Inspect 3D Nodes** - which meshes an option shows (auto-filled from your clusters; verify with Inspect).
- **Required Step** - must be answered before quoting.

Part D - Pricing (how a quote total is built)

The engine sums, in this order, and shows a live **Pricing Breakdown / Config Total**:

1. **Base assembly price** - the assembly's base price (or the flow's **Base Price** if no assembly).
2. **For each step:**
 - **Step Base Price** (if set) - set under **Item Mapping & Base Price**; the **Fetch** button pulls the linked part's base price automatically.
 - **+ Option native price** - the selected part's own base price (for Choose/Swap, the per-option **price** you set in Style Options wins).
 - **+ Upcharge** - the per-option **Upcharge (\$)** in the Option Properties table (**priceMap**).
 - **$\times$ quantity** (and any multiplier).
3. **Flat Price Override (\$)** - under **Pricing Rules**, if set, *replaces* that step's whole price (ignores native + upcharge). Use to force a fixed price.

4. **Static fees** - STATIC_FEE steps add their amount $\times$ qty.

Rule of thumb: per step the price is **Override (if set)**, otherwise **Base + Native + Upcharge**, then $\times$ qty. Set component prices from the BOM by using **Fetch**, and use **Upcharge** for "this option costs more than the default," and **Override** only when you want to ignore everything else.

Part E - Customer pricing (private-label / per-client)

1. In the CPQ run screen, use the top "**Active Customer:**" dropdown ("-- Select Customer to Activate Live Client Pricing --"). Until a customer is chosen, the customer sees **base MSRP**.
2. On each part you can store **client pricing** (the customer's own SKU and their price). For a step to use it, turn on "**Enable Client-Specific Pricing**" (useClientPricing) in the step's **Pricing Rules**.
3. With a customer active + client pricing enabled + a matching entry on the part, that step prices at the **customer's price** instead of the base price.

Resolution order: **Flat Price Override** beats everything -> otherwise **client price** (when customer + client pricing on) -> otherwise **base/native price**. Upcharges still add unless an override is set. (Note: the engine currently prices off the client price; client SKU is stored for paperwork.)

Part F - Verify it works

1. Open **Tab 8 CPQ Configurator, Step 1: Select Flow**, pick your flow.
 2. The model should render a complete default configuration. Step through; each pick should swap the right part in 3D (use the option's geometry to confirm).
 3. Pick a customer and confirm the price shifts to client pricing where you enabled it.
 4. If a part shows in the wrong place or not at all, it's a **grouping** issue - go back to Node Grouping and use **Locate** to find the bad cluster.
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Quick worked examples

- **Flat Iron (M2C):** Auto-Group -> Split by Position -> From Library -> Save -> Locate-check the center bracket -> Generate Flow from Tags -> set prices via Fetch + Upcharges -> test in CPQ.
- **Brimar (Classical Elements):** same, but **toggle Flip L/R** in Auto-Group first (its model is mirrored), then proceed identically. Verify with Locate that LEFT is the on-screen left before generating the flow.